UNIVERSITY OF TECHNOLOGY, SYDNEY

Data Science Internship Proposal

Internship Plan Report

Professional Internship Plan

Synogize Dealership Data Enablement Project

Snowflake Medallion Architecture, BI, and Analytics Use Cases

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1 Internship Plan Overview

This internship is undertaken with Synogize, a Data & AI consulting company that designs and delivers analytics, AI, and data platform solutions for business problems. The data context comes from Pentana ERA Power, a dealership management system used in the automotive industry. This operational system captures dealership processes across areas such as vehicle sales, inventory, finance, and customer operations, which makes it highly relevant for analytics use cases. Published reporting described eraPower as being connected to around 2,500 new-car franchise dealerships, or approximately 60% of the Australian market, and more recent internal discussion suggests the footprint may now be closer to 80%. This reinforces the commercial relevance of the data for dealer-focused analytics (Barry, 2020; Openpay Group Limited, 2020).

The project focuses on making this dealership data asset useful for business and analytics. Initial exploratory data analysis has already been completed on the raw dealership sales data extracted from the operational system. The current working dataset is substantial, with approximately 8.7 million rows and 143 columns, which makes data quality, structure, and modelling especially important. That work identified key variables, quality issues, and the likely analytical value of the data. This plan therefore focuses on what comes next: transforming the raw environment into a structured, reusable, and business-ready data foundation.

The main technical direction is a Snowflake medallion architecture. The Bronze layer will preserve the raw dealership data, the Silver layer will apply cleaning and standardisation, and the Gold layer will produce business-ready datasets for reporting and dashboarding. Given the size and complexity of the source data, the Silver layer is not only intended for cleaning; it is also where the main raw table will be split into more useful analytical structures that better support downstream reporting and modelling. This approach supports governance, traceability, and scalable analytics development. It also gives Synogize a clear pathway for showing how operational DMS data can support dealer-facing use cases.

The internship combines data engineering, analytics strategy, and business discovery. After cleaning and structuring the data, the next stage is to identify practical business use cases and BI opportunities. These may include dealership sales performance tracking, stock and model mix visibility, finance and profitability reporting, and future analytics or AI use cases. The main variable groups that support this work are summarised in Appendix A. In this sense, the goal is to transform raw data from a dealership management system into data that can fuel insights for dealerships.

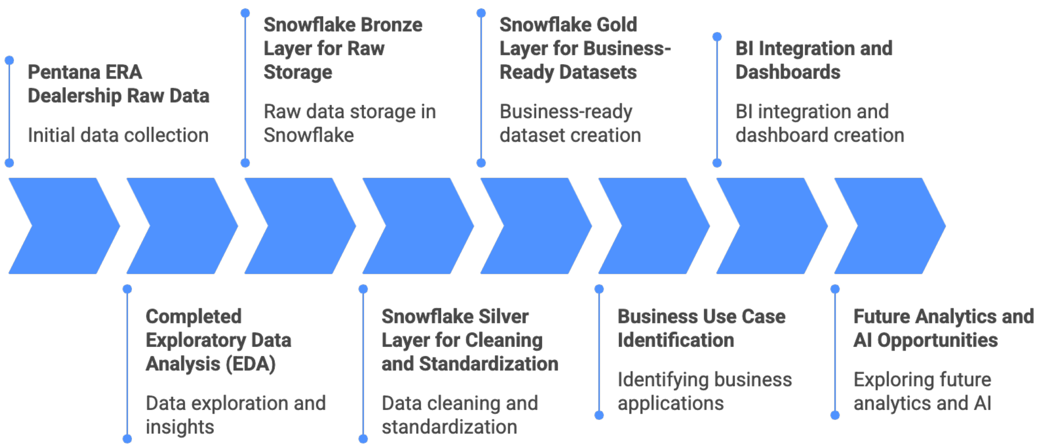

Figure 1: Planned internship workflow from raw Pentana data to business use cases

Overall, this internship is a plan to turn raw dealership data into a useful analytics asset within a real consulting environment. It starts from completed EDA, moves through Snowflake Bronze, Silver, and Gold layers, and then focuses on use cases, BI integration, and future analytics value for dealers.

2 Motivation and Project Summary

My motivation for undertaking this internship is closely connected to my professional transition into the data field and my long-term career direction. I originally trained and worked as a petroleum engineer, but developed a strong interest in data, analytics, and the use of information to support better decisions. That interest led me to complete a postgraduate qualification in analytics, work in data-related roles, and continue into the Master of Data Science and Innovation.

In previous roles I worked across both data engineering and data science, and over time I found that my strongest interest was in analytics and business value creation through data. For that reason, an internship with Synogize is especially attractive because it connects data engineering, analytics, BI, and AI to business outcomes.

This internship is also important to me because it provides an opportunity to work on real company problems in an Australian professional context. As I approach graduation, I want to strengthen my local professional experience and contribute to projects that solve practical business problems rather than only academic exercises.

From a technical perspective, the project aligns strongly with my interest in data engineering, analytics pipelines, and downstream analytical applications. Designing a Snowflake medallion architecture and improving the data through Bronze, Silver, and Gold layers will strengthen my practical skills in SQL transformation, cloud processing, data modelling, and analytics engineering.

In summary, the project objective is to make the dealership database useful. Using the completed EDA as a foundation, I will help define a Snowflake medallion architecture, improve the structure and quality of the data, and identify business use cases and BI integration opportunities that Synogize can explore further.

3 Project Requirements, Assessment Tasks, and Criteria

The internship will be structured through milestones across the semester so that the work remains organised and assessable. Each milestone corresponds to a specific phase of the project and contributes to the final analytics plan.

Table 1: Semester milestone and assessment plan

Week	Task	Weight
Week 1	Internship onboarding, project familiarisation, and review of Synogize, Pentana, and the ERA Power data context	0%
Week 2	Initial review of available documentation, business context, and previously completed exploratory analysis outputs	0%
Week 3	Confirmation of project scope, milestone planning, and alignment with supervisor expectations for the semester	0%
Week 4	Consolidation of completed exploratory data analysis, including key findings, data quality issues, and implications for architecture design	10%
Week 5	Design of the Snowflake medallion architecture across Bronze, Silver, and Gold layers	10%
Week 6	Implementation plan for ingestion and Silver-layer cleaning logic, including standardisation, deduplication, and validation checks	20%
Week 7	Refinement of data quality strategy through AI-assisted or rule-based approaches	10%
Week 8	Design of Gold-layer business-ready datasets and preparation for BI integration	10%
Week 9	Identification and prioritisation of business use cases supported by the curated database	10%
Week 10	Evaluation of BI integration options and future analytics opportunities	10%
Week 11	Verbal presentation to the Synogize industry supervisor on architecture, cleaning strategy, and use cases	10%
Week 12	Final internship report and technical documentation	10%

The first milestone formalises the completed EDA as the starting point for the project. The second milestone converts that understanding into a Snowflake medallion design.

The most heavily weighted milestone focuses on ingestion and Silver-layer cleaning logic because this stage creates the foundation for every later analytical outcome. The data quality refinement milestone then extends this work through rule-based or AI-assisted standardisation.

The later milestones shift the project from data preparation into business usefulness. Gold-layer design, BI integration planning, and business use-case identification show how the cleaned database can support reporting and value creation. The final presentation and report assess how clearly I can communicate the architecture, analytical opportunities, and industry relevance of the work.

4 Internship Timeframe, Meetings, and Milestones

The internship will run from 24 February 2026 to 29 May 2026, with an expected total workload of approximately 196 hours across the semester. Internship work is primarily completed through office attendance every Tuesday from 9:00 a.m. to 5:00 p.m., with additional work completed from home on another day depending on the weekly schedule and project priorities.

Regular engagement with the industry supervisor is central to project governance. Meetings with William So, Head of AI & Analytics at Synogize, are normally held on Tuesdays, with additional meetings scheduled when needed. The internship also includes discussions with automotive business experts and technical specialists within Synogize to connect the work to both business relevance and implementation practice.

The internship is being completed within a group environment involving four interns. Some parts of the work are shared across the team, including discussion of findings, joint problem solving, and project decisions around data structure, business use cases, and technical priorities.

Expected deliverables will follow the milestone structure. Early outputs focus on the EDA summary and architecture design, mid-semester outputs focus on cleaning logic and business-ready datasets, and later outputs focus on use cases, BI concepts, and the final report.

Progress will be tracked through milestone completion, supervisor feedback, meeting notes, and documented outputs in the project environment. This keeps the internship measurable across the semester and supports coordination across the intern team.

5 Project Objectives and Approach

The internship has four primary objectives:

- Make Pentana's dealership sales database useful for analytics and business decision-making.
- Build a Snowflake medallion architecture across Bronze, Silver, and Gold layers.
- Clean, restructure, and prepare the data for analytics and BI use.
- Identify practical business use cases and future analytics opportunities.

To achieve these objectives, the project will follow a staged technical approach guided by the EDA already completed on the ERA Power dealership data. The Snowflake architecture will retain raw data in Bronze, apply cleaning and restructuring in Silver, and curate business-facing datasets in Gold.

SQL-based extraction and transformation will be the main mechanism for data preparation. Silver will apply standardisation, null handling, duplicate management, category reconciliation, and validation of important dates and numeric measures. Because the dataset is large and wide, with approximately 8.7 million rows and 143 columns, this stage will also involve splitting the main raw table into more suitable analytical tables for downstream use.

The project will also assess whether AI-assisted methods can improve data quality through tasks such as category harmonisation, anomaly detection, and enrichment of poorly structured values.

Once the Gold layer has been defined, the project will move into business use-case discovery and BI enablement, including identification of relevant dashboards, reports, analytical questions, and future predictive opportunities.

6 Achieving Subject Learning Outcomes (SLOs)

This internship contributes directly to the subject learning outcomes published in the UTS handbook for the Data Science internship subject.

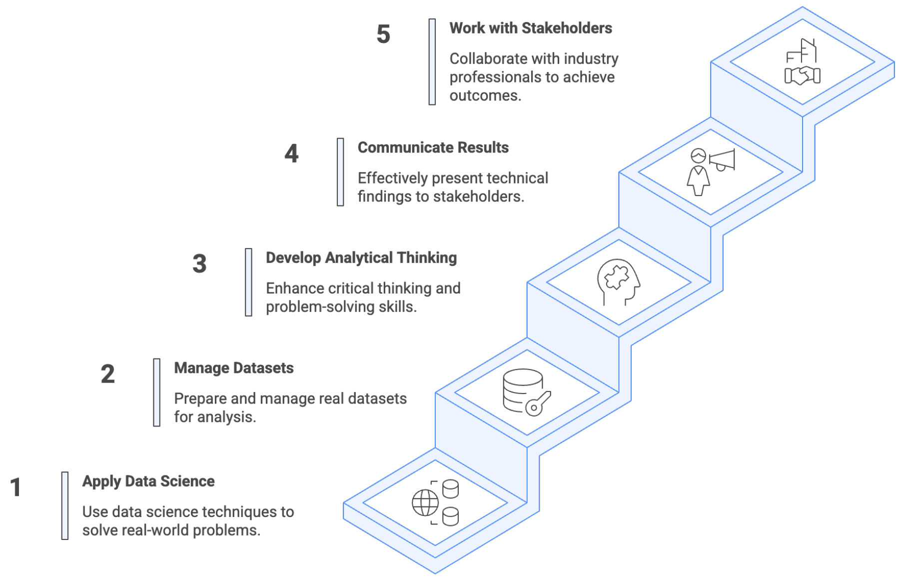

Figure 2: Internship activities aligned to subject learning outcomes

Figure 2 shows how the internship activities align with the subject learning outcomes. The project supports the application of data science knowledge in a workplace through work on real dealership data, Snowflake architecture, BI planning, and analytics design. It also supports productive contribution to organisational data science projects through collaboration with the intern team and Synogize staff.

The internship further develops professional communication, stakeholder engagement, and workplace behaviour through regular meetings, milestone reviews, collaboration with automotive business experts, and reporting to the supervisor. Reflection and communication of internship outcomes are supported through the final report and presentation of project results.

Overall, the internship aligns well with the official subject learning outcomes while building practical experience in cloud data engineering, analytics consulting, and business-oriented solution design.

7 Supervisor Agreement Section

"I agree with the items 3 and 4 outlined by the student."

Industry Supervisor Name: William So

Signature: ____

Date: ____

A Appendix: Bronze EDA Variable Groups

The internship uses the variable grouping developed during the Bronze EDA phase and documented in the project repository. These groups help connect raw dealership fields to business use cases and later Silver and Gold layer design.

Table 2: Summary of major variable groups from Bronze EDA

Variable Group	Cols	Business Relevance
Vehicle identity and registry	3	Vehicle-level tracking, inventory linking, and entity resolution
Deal and transaction identity	4	Quote, deal, and invoice linkage across the sales process
Vehicle descriptors and specifications	16	Model mix, segmentation, product analysis, and trend reporting
Dealer and factory option packages	39	Option uptake, configuration analysis, and pricing enrichment
Geography and delivery location	4	Regional demand, delivery footprint, and dealer market analysis
Lifecycle dates and process milestones	16	Operational timeline analysis, delays, and conversion tracking
Core commercial amounts	8	Pricing, cost, revenue, and margin analysis
Charge components and adjustments	16	Discount analysis, on-road cost breakdown, and pricing transparency
Aggregated totals and profitability metrics	19	Gross profit analysis and customer total visibility
Finance and F&I fields	9	Finance penetration, finance revenue, and contract-level insights
Sales channel, flags, and export metadata	5	Sales type analysis, channel reconciliation, and operational controls

The detailed category mapping and field-level definitions are maintained in the Bronze EDA notebook and project documentation. In this report, the appendix is included only as a reference point for the structure of the raw data and the design of later business-facing datasets.

References

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